

# Hypothesis Testing with E-Values

## Chapter 1: Introduction

Based on Ramdas & Wang

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# Chapter Overview

- 1 Notation you'll need
- 2 1.1 The three roles of e-values
- 3 1.2 P-values versus e-values
- 4 1.3 Mathematical notation and conventions
- 5 1.4 Definitions: e-variables, p-variables, tests
- 6 1.5 Betting interpretation of e-values
- 7 1.6 A first example: likelihood ratio
- 8 1.7 A second example: the soft-rank e-value
- 9 1.8 Historical context
- 10 1.9 Road map
- 11 Summary

# Notation You'll Need for This Chapter I

Before diving in, here is a plain-English glossary of every non-elementary term and symbol used later in this deck. Nothing here requires measure theory — treat each analogy as the working definition you need.

- **Null hypothesis,  $\mathcal{P}$ :** the “boring” explanation of the data we test against – a *set* of candidate probability distributions (e.g., “the coin is fair,” i.e.,  $\{\text{Bernoulli}(0.5)\}$ ).
- **Alternative hypothesis,  $\mathcal{Q}$ :** the “interesting” explanation we’d like evidence for (e.g., “the coin is biased towards heads”).
- $\mathbb{P}, \mathbb{Q}$ : a single probability distribution;  $\mathbb{P} \in \mathcal{P}$  means  $\mathbb{P}$  is one specific member of the null hypothesis set.
- **Simple vs. composite hypothesis:** simple means the set has exactly one distribution (e.g.,  $\{\mathbb{P}\}$ ); composite means it has more than one (e.g., “ $\theta \neq 0.5$ ” covers infinitely many biased coins).
- **Random variable  $X$ :** a quantity whose value depends on the random outcome (here: our data).
- $\mathbb{E}^{\mathbb{P}}[X]$ : the expectation (long-run average) of  $X$ , computed *assuming*  $\mathbb{P}$  is the true distribution.

# Notation You'll Need for This Chapter II

- **E-variable / e-value**: a nonnegative random variable  $E$  with  $\mathbb{E}^{\mathbb{P}}[E] \leq 1$  under every null distribution. Analogy: a bet that is never favorable to you (on average) if the null is true, but pays off big if the null is false. “E-value” is the number you actually observe.
- **P-variable / p-value**: you already know this one — a number in  $[0, 1]$  that is uniform-or-larger under the null. Small p-value = evidence against the null. (*Opposite* convention to e-values: for e-values, *large* is evidence against the null.)
- **Test**  $\phi$ : a (possibly randomized) decision rule taking values in  $[0, 1]$ , interpreted as “the probability I reject  $H_0$ .” A **binary** test only ever outputs 0 (“don’t reject”) or 1 (“reject”).
- **Type-I error / level**  $\alpha$ : the chance of a false alarm (rejecting a true null). A “level- $\alpha$  test” controls this chance to be at most  $\alpha$  (e.g.,  $\alpha = 0.05$ ).
- **Likelihood ratio**  $d\mathbb{Q}/d\mathbb{P}$ : for a data point  $x$ , how much more likely  $x$  is under  $\mathbb{Q}$  than under  $\mathbb{P}$ . If data are continuous/discrete with densities (or probability mass functions)  $q(x), p(x)$ , this is just  $q(x)/p(x)$ .

## Notation You'll Need for This Chapter III

- **Exact e/p-variable:** one where the defining inequality is always an equality ( $\mathbb{E}^{\mathbb{P}}[E] = 1$ , or  $\mathbb{P}(P \leq \alpha) = \alpha$ ) — the “tightest possible” case.
- **Stochastically larger:** random variable  $X$  is stochastically larger than a reference distribution if  $X$  is, loosely, “at least as likely to be big.” (Formally: its cdf is never above the reference cdf.)
- **Supermartingale idea (informal):** a wealth process that, on average, never grows from one step to the next given the past — like a casino game that is fair or tilted against you. We will only need the intuition, not the formal machinery.
- **Filtration / adapted / stopping time (informal):** think of a filtration as “the information you’ve seen up to now, growing over time”; a process is adapted if, at each time, it only uses information seen so far; a stopping time is a rule for “when to stop” that only looks at data observed up to that moment (no peeking into the future).

# Notation You'll Need for This Chapter IV

- **E-process:** a sequence of e-values  $E_1, E_2, \dots$  (built up as data arrive) that remains a valid e-value *even if you stop early*, at any data-dependent stopping time. This is what lets e-values handle “peeking” safely.
- **Exchangeable:** a collection of random variables whose joint behavior does not change if you relabel/permute them – e.g., shuffling a deck of otherwise-identical-looking cards.
- **Calibrator:** a fixed recipe that converts one kind of statistic into another while keeping validity, e.g., turning an e-value  $E$  into a p-value via  $P = 1/E$ .
- **Log-optimal e-value / e-power:** among all valid e-values, the one that maximizes  $\mathbb{E}^{\mathbb{Q}}[\log E]$  (expected log growth rate under the alternative  $\mathbb{Q}$ ) – explained with concrete intuition in Section 1.2.
- **Markov's inequality (used implicitly):** for any nonnegative random variable  $E$  with  $\mathbb{E}[E] \leq 1$ , we automatically get  $\mathbb{P}(E \geq 1/\alpha) \leq \alpha$  – this is exactly why “reject when  $E > 1/\alpha$ ” is a valid level- $\alpha$  test.

## 1.1 The Three Roles of E-Values I

An e-value (nonnegative statistic with expectation at most 1 under the null; reminder from the notation frame) plays three distinct roles in the book. Understanding these upfront will help you see why so many later chapters revisit the same simple object.

### Defining property (restated)

An e-value is a nonnegative test statistic whose expected value is at most one under the null hypothesis.

- “e” can stand for *expectation* (its defining constraint) or *evidence* (how we use it).
- Larger e-value  $\Rightarrow$  more evidence *against* the null – the opposite direction from p-values.

## 1.1 The Three Roles of E-Values II

- ① **Methodological.** E-values directly yield familiar tools: the reciprocal  $1/E$  of an e-value is a valid p-value, and rejecting the null when  $E$  exceeds  $1/\alpha$  is a valid level- $\alpha$  test (both via Markov's inequality). E-values also unlock things p-values struggle with: universal inference for irregular problems, log-optimal (“uniformly most e-powerful”) e-values, sequential anytime-valid testing, and simple false-discovery-rate control.
- ② **Technical.** E-values show up as intermediate tools even in problems that don't look like they involve e-values at all – e.g., every procedure that controls the false discovery rate turns out to be an instance of an e-value-based procedure in disguise. They are also intimately tied to nonnegative (super)martingales (the “fortune process never grows on average” idea from the notation frame).
- ③ **Fundamental.** At the deepest level: a powered test exists for a testing problem if and only if a powered e-variable exists. E-values are not just a convenience – they are equivalent, foundational objects to hypothesis tests themselves.

The book positions itself as building *a theory of evidence* (via e-values), complementing the classical *theory of decision-making* (Neyman, Pearson, Wald, Savage).



## 1.2.1 P-values and E-values Are Statistical Cousins I

The book does *not* argue e-values should replace p-values. Instead, it argues both deserve equal footing as complementary tools. Why call them “cousins”? Because they can be converted into each other while preserving key guarantees.

## 1.2.1 P-values and E-values Are Statistical Cousins II

### E-to-p and p-to-e calibration

The intuition: an e-value's reciprocal behaves like a p-value, and an indicator of “was the p-value small” behaves like an e-value.

$$\text{(e-to-p)} \quad P = \frac{1}{E} \qquad \text{(p-to-e, at level } \alpha) \quad E = \frac{\mathbf{1}\{P \leq \alpha\}}{\alpha}.$$

- $E$ : an e-variable for  $\mathcal{P}$ ;  $P$ : a p-variable for  $\mathcal{P}$ .
- $\mathbf{1}\{P \leq \alpha\}$ : the indicator function, 1 if  $P \leq \alpha$  is true, else 0.
- e-to-p calibration *preserves the rejection decision exactly*: for any  $\alpha$ ,  $1/E \leq \alpha \iff E \geq 1/\alpha$ .
- The specific p-to-e calibrator above also preserves the decision at that fixed level  $\alpha$  (it is the “all-or-nothing” e-value).

## 1.2.1 P-values and E-values Are Statistical Cousins III

**Key differences** (why you might prefer one over the other):

- **Post-hoc decisions.** With e-values, the rejection level  $\alpha$  can be chosen *after* seeing the data, and the guarantee still holds. This is provably false for p-values in general.
- **Optional continuation.** If several scientists sequentially run follow-up experiments, their e-values can always be *merged by simple multiplication* – even if later experiments were only run because earlier ones looked promising. Sequentially-obtained p-values can be very hard to combine correctly in this setting.
- **Unknown sampling scheme.** P-values need a fully specified data-collection design to be valid; a specific family of e-values remains valid regardless of the (reasonable) stopping/design rule used – illustrated concretely in Example 1.4 (Section 1.6).
- **Multiple testing.** “Compound” e-values turn out to be central to multiple-testing procedures (Chapters 9, 11, 13) in ways that have no clean p-value analogue.

## 1.2.2 Power and E-power I

A natural question: *are e-values more or less powerful than p-values?* The book's answer is subtle – power is a property of a *procedure* (a threshold test), not of e-values or p-values in isolation.

- If a p-value  $P$  is derived from an e-value  $E$  via  $P = 1/E$ , the two thresholded tests are identical – no difference in power.
- If an e-value is derived from a p-value via  $E = 1\{P \leq \alpha\}/\alpha$ , the resulting level- $\alpha$  test again has exactly the same power as the original p-value test.
- Otherwise (general e-values vs. general p-values), no universal comparison is possible – it depends on which specific e-value or p-value you picked.

## 1.2.2 Power and E-power II

### E-power: the criterion the book actually optimizes

Intuition first: if we care about the null hypothesis being *wrong* (alternative  $\mathbb{Q}$  true), we'd like our e-value to grow as fast as possible under  $\mathbb{Q}$ . Taking the expected *logarithm* (rather than the expectation) turns out to be the right notion of “fast growth” when e-values will be multiplied together across repeated experiments.

$$\text{e-power of } E \text{ against } \mathbb{Q} := \mathbb{E}^{\mathbb{Q}}[\log(E)].$$

- $\mathbb{E}^{\mathbb{Q}}[\cdot]$ : expectation computed as if the alternative  $\mathbb{Q}$  generated the data.
- $\log$ : natural logarithm.
- An e-value that maximizes e-power is called **log-optimal**.

## 1.2.2 Power and E-power III

- Why the log? If scientists sequentially multiply e-values  $E_1, E_2, \dots$  (valid even under data-dependent stopping/continuation), then  $\prod_{i=1}^n E_i = \exp(n \cdot \overline{\log E}_{[n]})$ , so the *rate* at which the product diverges to  $\infty$  is governed exactly by  $\mathbb{E}[\log E_i]$ .
- Log-optimality is therefore not an arbitrary choice of utility function – it is forced by the structure of repeated multiplication.

## 1.2.2 Power and E-power IV

One subtlety worth remembering: e-values in this book are chosen to be log-optimal, *not* to maximize power at a fixed level  $\alpha$ .

- A power-maximizing e-value degenerates into an “all-or-nothing” e-value  $\mathbf{1}_A/\alpha$  for some rejection set  $A$  – but this has e-power  $= -\infty$  (it equals zero with positive probability, and  $\log(0) = -\infty$ ).
- Log-optimal e-values, by contrast, are always strictly positive (with probability 1 under  $\mathbb{Q}$ ), so they never risk this catastrophic penalty.

## 1.2.2 Power and E-power V

- **Takeaway:** if you plug a log-optimal e-value directly into a decision procedure built around classical power, you may be disappointed – not because e-values are weak, but because you're comparing objects optimized for different criteria.



# Python: E-to-P and P-to-E Calibration, Checked Numerically

```
import numpy as np
rng = np.random.default_rng(0)

alpha = 0.05
n_reps = 200_000

# Start from a valid e-value E (mean <= 1 under H0) for our coin example:
# H0: theta=0.5 (fair coin), n=20 flips, LR e-value vs theta=0.7
n, p0, q0 = 20, 0.5, 0.7
S = rng.binomial(n, p0, size=n_reps) # data simulated under H0
E = (q0/p0)**S * ((1-q0)/(1-p0))**(n-S) # e-value (eq. 1.5)

P_from_E = 1.0 / E # e-to-p calibration
print("mean(E) under H0", E.mean()) # should be <= 1
print("P(P_from_E <= alpha)", (P_from_E <= alpha).mean(),
      "vs alpha =", alpha) # should be <= alpha
print("P(E >= 1/alpha)", (E >= 1/alpha).mean()) # identical event

# Now go the other way: turn a p-value into an e-value at level alpha
E_from_P = (P_from_E <= alpha).astype(float) / alpha # p-to-e calibration
print("mean(E_from_P) under H0 =", E_from_P.mean()) # should be <= 1
```

**Output (excerpt):**  $\text{mean}(E) \approx 0.982$ ,  $P(P\_from\_E \leq 0.05) \approx 0.006 \leq 0.05$ ,  $\text{mean}(E\_from\_P) \approx 0.006 \leq 1$  — exactly the two calibration guarantees from the block above, confirmed by simulation.

## 1.3 Mathematical Notation and Conventions I

This section fixes the formal scaffolding used throughout the book. We only need the parts relevant to the definitions in the next section – treat the rest as background color.

- Data live in a sample space  $\Omega$  (all possible outcomes), equipped with a  $\sigma$ -algebra  $\mathcal{F}$  (informally: “all the events we’re allowed to assign a probability to”; see notation frame).
- $\mathcal{M}_1$  = the set of *all* probability distributions on  $(\Omega, \mathcal{F})$ . Its elements are also just called “distributions.”
- A random variable  $X$  is a function from  $\Omega$  to  $\mathbb{R}$  (or  $[-\infty, \infty]$  if we explicitly allow infinite values).
- The true (but unknown) distribution generating our data  $X$  is  $\mathbb{P}^\dagger \in \mathcal{M}_1$ . Data can be a vector  $X = (X_1, \dots, X_n)$ , possibly (but not necessarily) i.i.d.
- $X^t := (X_1, \dots, X_t)$ : shorthand for the first  $t$  data points.

## 1.3 Mathematical Notation and Conventions II

### Definition 1.1: Hypothesis

*Intuition:* a hypothesis is simply the collection of distributions that are “consistent with” a scientific claim. If the claim pins down exactly one distribution, there’s no remaining uncertainty about which  $\mathbb{P}$  is true (simple); otherwise several distributions are all compatible with the claim (composite).

A *hypothesis* is a set of probability measures in  $\mathcal{M}_1$ . A hypothesis is *simple* if it is a singleton, such as  $\{\mathbb{P}\}$  and  $\{\mathbb{Q}\}$ . Otherwise it is *composite*.

- We always write  $\mathcal{P}$  for the null hypothesis and  $\mathcal{Q}$  for the alternative hypothesis;  $H_0$  /  $H_1$  denote the corresponding statements “ $\mathbb{P}^\dagger \in \mathcal{P}$ ” / “ $\mathbb{P}^\dagger \in \mathcal{Q}$ .”
- $\mathcal{P}, \mathcal{Q}$  are always assumed non-intersecting (they can’t both be true).
- $\text{Conv}(\mathcal{P})$ : the convex hull of  $\mathcal{P}$  (all probability-weighted mixtures of distributions in  $\mathcal{P}$ ).

## 1.3 Mathematical Notation and Conventions III

- $\mathbb{Q}$  is *absolutely continuous* with respect to  $\mathbb{P}$ , written  $\mathbb{Q} \ll \mathbb{P}$ , if  $\mathbb{P}(A) = 0 \Rightarrow \mathbb{Q}(A) = 0$  for every measurable set  $A$ . Intuition: anything  $\mathbb{P}$  deems impossible is also impossible under  $\mathbb{Q}$ , so it makes sense to compare their densities pointwise.
- When  $\mathbb{Q} \ll \mathbb{P}$ , we write  $d\mathbb{Q}/d\mathbb{P}$  for the likelihood ratio (reminder from the notation frame): if  $q, p$  are densities/pmfs of  $\mathbb{Q}, \mathbb{P}$ , then  $(d\mathbb{Q}/d\mathbb{P})(x) = q(x)/p(x)$ .
- $X$  is *stochastically larger* than a distribution with cdf  $F$  if  $\mathbb{P}(X \leq x) \leq F(x)$  for all  $x \in \mathbb{R}$  (reminder: “at least as likely to be big”).
- $\mathbb{N} = \{1, 2, \dots\}$ ;  $\mathbb{R} = (-\infty, \infty)$ ;  $[K] := \{1, \dots, K\}$ ;  $\Delta_K$  is the probability simplex in  $\mathbb{R}^K$ :  $\Delta_K = \{(v_1, \dots, v_K) \in [0, 1]^K : \sum_{k=1}^K v_k = 1\}$ .
- The Euler number  $\exp(1)$  is denoted  $e$  (upright) to avoid clashing with the variable  $E$  used for e-values.
- “Increasing” / “decreasing” are always meant in the *non-strict* sense;  $x \wedge y = \min(x, y)$ ,  $x \vee y = \max(x, y)$ .

## 1.4 Definitions of E-variables, P-variables, and Tests I

Here is the formal heart of the chapter. Before the wall of symbols: an e-variable is a “fair-or-worse bet against the null,” a p-variable is the classical p-value object, and a test is a (possibly randomized) accept/reject rule. Let  $\mathcal{P} \subseteq \mathcal{M}_1$  be the null hypothesis.

### Definition 1.2: E-variables, P-variables, and Tests

- (i) An *e-variable*  $E$  for  $\mathcal{P}$  is a  $[0, \infty]$ -valued random variable satisfying  $\mathbb{E}^{\mathbb{P}}[E] \leq 1$  for all  $\mathbb{P} \in \mathcal{P}$ . It is *exact* if  $\mathbb{E}^{\mathbb{P}}[E] = 1$  for all  $\mathbb{P} \in \mathcal{P}$ .
- (ii) A *p-variable*  $P$  for  $\mathcal{P}$  is a  $[0, \infty)$ -valued random variable satisfying  $\mathbb{P}(P \leq \alpha) \leq \alpha$  for all  $\alpha \in (0, 1)$  and all  $\mathbb{P} \in \mathcal{P}$ . It is *exact* if  $\mathbb{P}(P \leq \alpha) = \alpha$  for all  $\alpha, \mathbb{P}$ .
- (iii) A *test*  $\phi$  is a  $[0, 1]$ -valued random variable. It is *binary* if its range is  $\{0, 1\}$ . The *type-I error (rate)* of  $\phi$  for  $\mathbb{P}$  is  $\mathbb{E}^{\mathbb{P}}[\phi]$ . A test has *level*  $\alpha \in [0, 1]$  for  $\mathcal{P}$  if its type-I error is at most  $\alpha$  for every  $\mathbb{P} \in \mathcal{P}$ .

- $\mathfrak{E} = \mathfrak{E}(\mathcal{P})$ : the set of *all* e-variables for  $\mathcal{P}$ ;  $\mathfrak{U} = \mathfrak{U}(\mathcal{P})$ : the set of all p-variables for  $\mathcal{P}$ .

## 1.4 Definitions of E-variables, P-variables, and Tests II

### Remarks worth internalizing:

- 1 A test allows external randomization: draw  $U \sim \text{Unif}(0, 1)$  independent of the data, reject if  $U \leq \phi$ . Then  $\mathbb{E}^{\mathbb{P}}[\phi]$  is exactly the unconditional rejection probability.
- 2 Equivalently,  $P$  is a p-variable iff  $P$  is stochastically larger than  $\text{Unif}(0, 1)$  under every  $\mathbb{P} \in \mathcal{P}$ .
- 3 A large realized e-value is evidence *against*  $H_0$  (because its mean is capped at 1 under the null, so a big value is “surprising”); a small realized p-value is likewise evidence against  $H_0$ .
- 4 We say “an e-variable for  $H_0$ ” to mean “for  $\mathcal{P}$ .”
- 5 Most tests we use are binary, with  $\phi_0 \equiv 0$  (never reject) as the trivial level-0 test.
- 6 An e-variable is allowed to equal  $\infty$  (licence to reject outright), but only on *null-sets* (events impossible under every  $\mathbb{P} \in \mathcal{P}$ ) – otherwise  $\mathbb{E}^{\mathbb{P}}[E]$  couldn’t stay  $\leq 1$ . The constant  $E \equiv 1$  is always a valid (if useless) e-variable for any  $\mathcal{P}$ .

## 1.4 Definitions of E-variables, P-variables, and Tests III

### Fact 1.3: Useful closure properties

*Intuition:* these are the “legal moves” you’re allowed to make with e- and p-variables without breaking validity – e.g., you can always average several e-values together, or multiply *independent* ones, and the result is still a legitimate e-variable. Let  $F_{X|\mathbb{P}}$  be the cdf of  $X$  under  $\mathbb{P}$ .

- ❶ For any  $X, \mathbb{P}$ :  $F_{X|\mathbb{P}}(X)$  is a p-variable for  $\mathbb{P}$ .
- ❷  $\sup_{\mathbb{P} \in \mathcal{P}} F_{X|\mathbb{P}}(X)$  is a p-variable for  $\mathcal{P}$ .
- ❸ If  $E_1, \dots, E_K$  are e-variables for  $\mathcal{P}$ , their **arithmetic average** is an e-variable for  $\mathcal{P}$ .
- ❹ If  $E_1, \dots, E_K$  are e-variables for  $\mathcal{P}$  and **independent** under each  $\mathbb{P} \in \mathcal{P}$ , their **product** is an e-variable for  $\mathcal{P}$ .
- ❺ If  $P$  is a p-variable and  $P' \geq P$ , then  $P'$  is a p-variable.
- ❻ If  $E$  is an e-variable and  $0 \leq E' \leq E$ , then  $E'$  is an e-variable.

## 1.4 Definitions of E-variables, P-variables, and Tests IV

### E-process (informal definition)

*Intuition:* an e-process is a whole time series of e-values that stays trustworthy *even if you decide when to stop looking based on the data itself* – crucial for sequential science.

A finite or infinite sequence  $E_1, E_2, \dots$  of e-variables (adapted to an underlying filtration; reminder: “only uses information seen so far”) is called an *e-process* if, for every stopping time  $\tau$  (a rule for stopping that doesn’t peek into the future),  $E_\tau$  is again a valid e-value.

- This innocuous-looking definition hides a deep connection to nonnegative (super)martingales, explored fully in Chapter 7.
- It is exactly the property that makes “optional continuation” (Section 1.2.1) and “unknown sampling schemes” (Example 1.4) work.



## Python: Checking Fact 1.3 (Averaging and Multiplying E-values)

```
1 import numpy as np
2 rng = np.random.default_rng(1)
3 n, p0, n_reps = 20, 0.5, 100_000
4 S = rng.binomial(n, p0, size=n_reps)      # data simulated under H0
5
6 # Two e-variables for the SAME null, vs two different alternatives
7 E1 = (0.7/p0)**S * (0.3/(1-p0))**(n-S)    # LR e-value vs theta=0.7
8 E2 = (0.6/p0)**S * (0.4/(1-p0))**(n-S)    # LR e-value vs theta=0.6
9 E_avg = (E1 + E2) / 2                     # Fact 1.3(iii): average
10 print("mean(E_avg) =", E_avg.mean())      # <= 1 ?
11
12 # Product of e-variables from INDEPENDENT rounds (Fact 1.3(iv)):
13 Sa = rng.binomial(10, p0, size=n_reps)    # first 10 flips
14 Sb = rng.binomial(10, p0, size=n_reps)    # independent next 10 flips
15 Ea = (0.7/p0)**Sa * (0.3/(1-p0))**(10-Sa)
16 Eb = (0.7/p0)**Sb * (0.3/(1-p0))**(10-Sb)
17 print("mean(Ea*Eb) =", (Ea * Eb).mean())  # <= 1 ?
```

**Output:** both means come out close to 1 (typically 0.97–1.00) and never exceed it, confirming Fact 1.3(iii)–(iv).

## 1.5 Betting Interpretation of E-values

E-variables have a clean interpretation as *fair or unfavorable bets* against the null. This framing is the intuitive engine behind most of the book's constructions.

### The Forecaster/Skeptic game

*Intuition:* Skeptic is trying to make money betting against Forecaster's claim that  $\mathbb{P} \in \mathcal{P}$  describes the world. An e-variable is precisely a bet that Forecaster (a rational, expected-value-maximizing party) would be willing to accept.

- Skeptic pays \$1 up front and proposes a payout function  $E : \Omega \rightarrow \mathbb{R}_+$ : if outcome  $\omega$  occurs, Forecaster pays Skeptic  $E(\omega)$  dollars.
- Forecaster accepts the contract only when it is favorable to them – i.e., when  $\mathbb{E}^{\mathbb{P}}[E] \leq 1$  (Skeptic doesn't expect to profit if  $\mathbb{P}$  is really true). This is *exactly* the e-variable condition.
- If Skeptic secretly believes  $\mathbb{Q}$  is the truth, they will only offer this particular  $E$  if  $\mathbb{E}^{\mathbb{Q}}[E] > 1$  – i.e., they expect to come out ahead.

A large realized e-value  $E(\omega)$  = Skeptic made a lot of money = strong evidence that Forecaster's null was wrong.

## Contrast: The (Flawed) Betting Interpretation of P-values

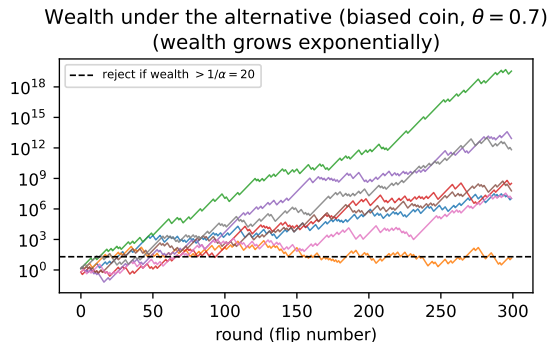
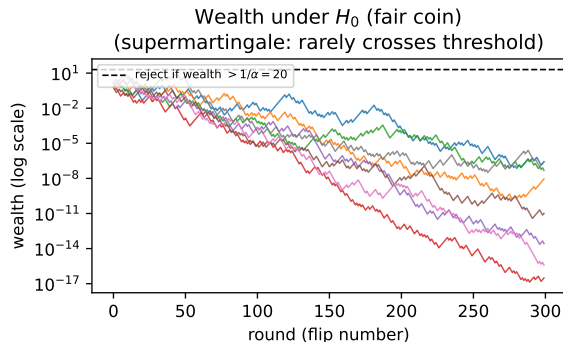
P-values have a betting interpretation too – but a flawed one, which helps explain why e-values are the more natural currency for combining evidence across experiments.

- For every  $r \in (0, 1]$ , Forecaster offers a contract priced at \$1 that pays  $1/r$  dollars if statistic  $P < r$ , and nothing otherwise.
- Forecaster only offers this if  $P$  is a p-variable for  $\mathcal{P}$  (equivalently,  $\mathbf{1}\{P \leq r\}/r$  is an e-variable for any fixed  $r$ ).
- But here, the bets are indexed by  $r$ , and the *realized* p-value equals the smallest  $r$  at which Skeptic *would have* profited, **had they picked that particular bet in advance**.
- This is “double dipping”: p-values implicitly report the best bet in hindsight, whereas an e-value is a single, honestly pre-committed bet. That is why e-values and  $1/(\text{p-value})$  should never be compared on the same numeric scale.

This distinction is exactly what powers the “optional continuation” and “post-hoc decision” advantages of Section 1.2.1.

# Figure: The Betting Wealth Process

Betting interpretation: Skeptic's wealth process, 8 sample paths



Eight simulated wealth paths from repeatedly betting  $E = (0.7/0.5)^{\text{heads}}(0.3/0.5)^{\text{tails}}$  per flip. **Left:** under the null ( $\theta = 0.5$ ), wealth fluctuates but rarely exceeds the rejection threshold  $1/\alpha$  (supermartingale intuition: on average, wealth does not grow). **Right:** under the true alternative ( $\theta = 0.7$ ), wealth grows exponentially – Skeptic profits.

# Python: Simulating the Wealth Process and Ville's Inequality

```
1 import numpy as np
2 rng = np.random.default_rng(2024)
3
4 alpha = 0.05
5 threshold = 1/alpha          # reject if wealth ever exceeds this
6 n_rounds = 300
7 n_reps = 5000
8 p0, q0 = 0.5, 0.7          # H0: fair coin; bet built against theta=0.7
9
10 max_wealth = np.zeros(n_reps)
11 for r in range(n_reps):
12     flips = rng.binomial(1, p0, size=n_rounds)          # data under H0
13     per_flip_e = np.where(flips == 1, q0/p0, (1-q0)/(1-p0))
14     wealth = np.cumprod(per_flip_e)                    # running product = wealth
15     max_wealth[r] = wealth.max()
16
17 frac_exceeds = np.mean(max_wealth > threshold)
18 print(f"P(max wealth over {n_rounds} rounds > 1/alpha={threshold:.0f}) = {frac_exceeds:.4f}")
19 print(f"Ville's inequality guarantees this is <= alpha = {alpha}")
```

**Output:**  $P(\max \text{ wealth} > 20) = 0.0490$ , comfortably respecting the  $\leq 0.05$  guarantee – even though we checked the *running maximum* over all 300 rounds, not just the final value. This is the practical payoff of the e-process property.

## 1.6 A First Example: Likelihood Ratio

The most natural e-variable of all: for testing a simple null  $\mathbb{P}$  against a simple alternative  $\mathbb{Q} \ll \mathbb{P}$  (reminder: absolutely continuous, i.e., comparable densities), the likelihood ratio itself is an e-variable.

### Likelihood-ratio e-variable

*Intuition:* how much more plausible is the data under  $\mathbb{Q}$  than under  $\mathbb{P}$ ? That ratio, applied to the observed data, is automatically a valid (exact) e-value for  $\mathbb{P}$ .

$$E = \frac{d\mathbb{Q}}{d\mathbb{P}}(X).$$

- $X$ : the observed data.
- For *any*  $\mathbb{Q} \ll \mathbb{P}$ , this  $E$  is an e-variable for  $\mathbb{P}$ , and in fact *exact* ( $\mathbb{E}^{\mathbb{P}}[E] = 1$ ).

We specialize to two settings the book uses throughout: Gaussian mean shift, and our running Bernoulli (coin flip) example.

## 1.6 Likelihood Ratio: the Gaussian Example

### Gaussian example (Eq. 1.1–1.2)

Testing  $\mathbb{P} = \mathcal{N}(0, 1)$  against  $\mathbb{Q} = \mathcal{N}(\mu, 1)$  ( $\mu > 0$  known), with i.i.d. data  $X_1, \dots, X_n$  and  $S_n = \sum_i X_i$ :

$$E_n = \exp(\mu S_n - n\mu^2/2) = \prod_{i=1}^n \exp(\mu X_i - \mu^2/2).$$

- $S_n$ : sum of the  $n$  observations.
- Each factor  $\exp(\mu X_i - \mu^2/2)$  is itself an exact e-variable; their product remains exact ( $\mathbb{E}^{\mathbb{P}}[E_n] = 1$ ).

Writing  $\delta = \mu\sqrt{n}$  (signal strength) and  $Z = S_n/\sqrt{n}$  (standard normal under  $H_0$ ):

$$E_n = \exp(\delta Z - \delta^2/2), \quad P_n = \Phi(-S_n/\sqrt{n}) = \Phi(-Z)$$

- $\Phi$ : standard normal cdf;  $P_n$  is the classical Neyman–Pearson p-variable.
- $E_n$  increasing in  $S_n$ ,  $P_n$  decreasing in  $S_n$ : both agree that larger  $S_n$  favors  $\mathbb{Q}$ .

## 1.6 Likelihood Ratio: the Two-Sided Version

### Two-sided version (Eq. 1.3–1.4)

Testing  $N(0, 1)$  against  $N(\mu, 1)$  for *unknown*  $\mu \neq 0$ , using a free parameter  $\delta > 0$ :

$$E_n = \frac{\exp(\delta Z - \delta^2/2) + \exp(-\delta Z - \delta^2/2)}{2}, \quad P_n = 2\Phi(-|Z|).$$

- Averaging the “ $+\mu$ ” and “ $-\mu$ ” bets (Fact 1.3(iii)) gives a valid e-variable for the two-sided problem.
- $\delta$  is a design choice affecting power – addressed in later chapters.



## 1.6 Likelihood Ratio: the Bernoulli (Coin-Flip) Version

### Bernoulli / coin-flip version (Eq. 1.5) – our running example

Testing  $\mathbb{P} = \text{Bernoulli}(p)$  vs.  $\mathbb{Q} = \text{Bernoulli}(q)$ ,  $q > p$ , sample size  $n$ ,  $S_n = \sum_i X_i$  (number of heads):

$$E_n = \left(\frac{q}{p}\right)^{S_n} \left(\frac{1-q}{1-p}\right)^{n-S_n}.$$

- $E_n$  exact, increasing in  $S_n$ ; Neyman–Pearson p-variable is  $P_n = 1 - F_{n,p}(S_n)$  (upper binomial tail).
- This is our **running example** for the rest of the deck: testing whether a coin is fair.

## Worked Example: Is Our Coin Fair? (Step by Step)

**Setup.**  $H_0 : \theta = 0.5$  (fair coin) vs. chosen alternative  $\theta = 0.7$ . We flip  $n = 20$  times and observe  $S_{20} = 14$  heads.

### Computing the e-value, one step at a time

- **Step 1** (identify the formula): use Eq. 1.5 with  $p = 0.5$ ,  $q = 0.7$ ,  $n = 20$ ,  $S_n = 14$ .
- **Step 2** (plug in):  $E_{20} = (0.7/0.5)^{14} (0.3/0.5)^6 = (1.4)^{14}(0.6)^6$ .
- **Step 3** (compute each factor):  $1.4^{14} \approx 111.12$ ,  $0.6^6 = 0.046656$ .
- **Step 4** (multiply):  $E_{20} \approx 111.12 \times 0.046656 \approx 5.18$ .
- **Step 5** (interpret):  $E_{20} \approx 5.18 > 1$  is mild evidence *against* the fair-coin null – but well below  $1/\alpha = 20$  for  $\alpha = 0.05$ , so we would *not* reject at that level.

**For comparison**, the exact one-sided p-value is  $P(S_{20} \geq 14 \mid \theta = 0.5) \approx 0.058$  – also not significant at  $\alpha = 0.05$ , consistent with the e-value's verdict.

## Example 1.4: Unknown Sampling Scheme I

This is the book's own example illustrating why e-values tolerate unknown data-collection designs, while p-values do not.

### Setup

Testing  $\mathbb{P} = \text{Bernoulli}(1/2)$  vs.  $\mathbb{Q} = \text{Bernoulli}(q)$ ,  $q > 1/2$ . A scientist is handed the dataset  $(1, 1, 0, 1, 1, 1, 1, 1)$  – 8 points, 7 heads – *without* being told how it was collected.

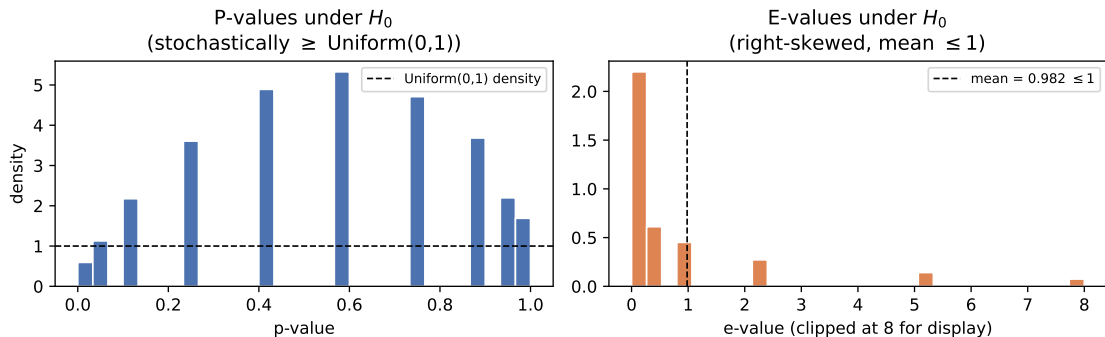
- **Design (a):** collect exactly 8 points. P-value  $= \mathbb{P}(N_8 \geq 7) = 0.035$  (significant at 0.05).
- **Design (b):** collect data until 5 heads occur in a row. P-value  $= \mathbb{P}(5\text{-in-a-row observed at or before } n = 8) = 0.078$  (*not* significant at 0.05).

## Example 1.4: Unknown Sampling Scheme II

- **The problem:** if only the data (not the design) is reported, the scientist cannot tell whether they're in case (a), (b), or some other design – yet the p-value's validity crucially depends on knowing which.
- **The fix:** the e-value  $E_n = q^{S_n}(1 - q)^{n - S_n}/2^n$  from Eq. 1.5 (with  $p = 1/2$ ) remains a *valid* e-variable **regardless of the data-collecting design**, as long as all obtained data is used from the start.
- **Why:**  $(E_n)_{n \geq 1}$  forms an e-process (Chapter 7) – it stays valid at any data-dependent stopping rule, which is exactly the “no peeking into the future” property from the notation frame.
- **Caveat:** this does not rescue *malicious* designs (cherry-picking only 1s, discarding early 0s, reporting only the “best” of several batches) – no statistical method purely based on collected data can fix that.

# Figure: P-values vs. E-values Under Repeated Null Experiments

Same 20-flip experiment, repeated 20{,}000 times under  $H_0$ : p-value vs. e-value



20,000 simulated repeats of our  $n = 20$  coin-flip experiment under  $H_0 : \theta = 0.5$ . **Left:** p-values are (stochastically) uniform on  $[0, 1]$  – as required by the p-variable definition. **Right:** e-values are right-skewed (most realizations are modest or small, a few are large) with sample mean  $\approx 0.98 \leq 1$ , exactly as the e-variable definition demands. Numerically,  $\mathbb{P}(E > 20) \approx 0.006 \leq 1/20$ , confirming Markov's inequality.

# An Illustration: Sequential Data and Adaptive Strategies I

Suppose data  $X_1, \dots, X_n$  from parameter  $\theta^\dagger$  arrive one at a time, testing  $H_0 : \theta^\dagger = 0$  vs.  $H_1 : \theta^\dagger \in \Theta_1$ . Let  $\ell(x; \theta) = d\mathbb{Q}_\theta/d\mathbb{P}(x)$ . Any choice of  $\theta$  turns  $\ell(X_i; \theta)$  into a valid e-variable for  $H_0$  – so we have freedom in how to pick  $\theta$  at each step:

- (a) **Fixed:** pick  $\theta_1, \dots, \theta_n \in \Theta_1$  in advance (possibly all equal).
- (b) **Adaptive:** estimate  $\theta_k$  from  $X_1, \dots, X_{k-1}$  only – e.g., via maximum likelihood (MLE) or maximum a posteriori (MAP).

# An Illustration: Sequential Data and Adaptive Strategies II

## The running product is always a valid martingale

*Intuition:* no matter how cleverly (or adaptively) you pick each  $\theta_i$ , as long as you only use *past* data to do so, multiplying per-round e-variables together keeps you honest – your expected next-step wealth, given everything so far, is exactly your current wealth.

Define  $M_n := \prod_{i=1}^n E_i$  with  $E_i = \ell(X_i; \theta_i)$ ,  $M_0 = 1$ .

$$\mathbb{E}^{\mathbb{P}}[M_i \mid X_1, \dots, X_{i-1}] = M_{i-1} \cdot \mathbb{E}^{\mathbb{P}}[\ell(X_i; \theta_i) \mid X_1, \dots, X_{i-1}] = M_{i-1}.$$

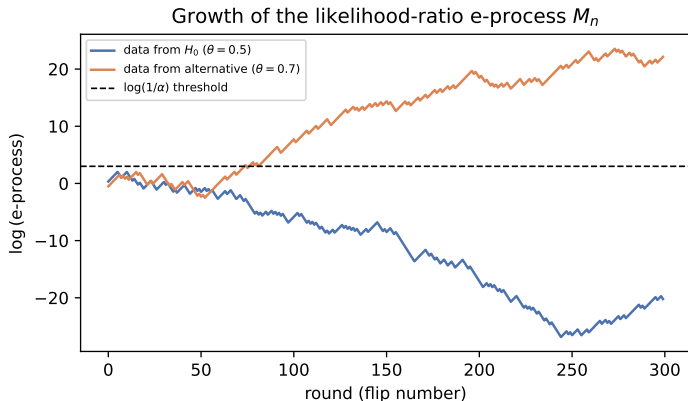
- **Step 1:** factor  $M_i = M_{i-1} \cdot E_i$  out of the conditional expectation ( $M_{i-1}$  is already known given the past).
- **Step 2:**  $\theta_i$  is computed from  $X_1, \dots, X_{i-1}$  only, so  $\mathbb{E}^{\mathbb{P}}[\ell(X_i; \theta_i) \mid X_1, \dots, X_{i-1}] = 1$  (it's an exact e-variable given the past).
- **Conclusion:**  $(M_i)$  never grows *on average* given the past – the martingale/supermartingale property from the notation frame.

## An Illustration: Sequential Data and Adaptive Strategies III

**Example 1.5 (book):** data from  $N(\theta^\dagger, 1)$  with true  $\theta^\dagger = 0.3$ ;  $H_0 : \theta^\dagger = 0$ . Six strategies for choosing  $\theta_i$ : (i) true value 0.3 (log-optimal); (ii) misspecified 0.1; (iii) random,  $\theta_i \sim \text{Unif}[0, 0.5]$  i.i.d.; (iv) MAP with prior  $N(0.1, 0.2^2)$ ; (v) MLE from  $X_1, \dots, X_{i-1}$  (started at  $\theta_1 = 0.1$ ); (vi) discrete mixture over a uniform grid on  $[0, 0.6]$ . For large  $n$ , methods (iv), (v), (vi) grow *almost as fast* as the (unrealistic, oracle) method (i); misspecified/random strategies (ii), (iii) grow more slowly.



# An Illustration: Sequential Data and Adaptive Strategies IV



Our own simplified illustration in the same spirit: a single e-process run under  $H_0$  (bounded, rarely crosses the  $1/\alpha$  threshold) versus under the true alternative (grows essentially linearly in log-scale, i.e. exponentially).

# Python: 1000 Replications — Confirming Mean E-value $\leq 1$

```
1 import numpy as np
2 rng = np.random.default_rng(3)
3
4 n, p0, q0 = 20, 0.5, 0.7
5 n_reps = 1000
6 S = rng.binomial(n, p0, size=n_reps)           # simulate data under H0, 1000 times
7 E = (q0/p0)**S * ((1-q0)/(1-p0))**(n-S)        # e-value for each replication
8
9 print("Sample mean of E over 1000 replications:", E.mean())
10 print("Sample max of E:", E.max())
11 print("Fraction with E > 1/0.05 = 20:", np.mean(E > 20))
12
13 # The one specific dataset we highlighted in the worked example:
14 S_obs = 14
15 E_obs = (q0/p0)**S_obs * ((1-q0)/(1-p0))**(n-S_obs)
16 print("E for our observed 14-heads-out-of-20 dataset:", round(E_obs, 4))
```

**Output (excerpt):** Sample mean of  $E \approx 0.99$  (consistent with the  $\leq 1$  guarantee up to Monte Carlo error), Fraction with  $E > 20 \approx 0.005$ -- $0.01$  ( $\leq 0.05$ , as Markov's inequality demands), and  $E$  for our observed dataset = 5.1844, matching the hand computation on the previous slide.

## 1.7 A Second Example: The Soft-Rank E-value

Many classical procedures (permutation tests, Monte Carlo tests) produce statistics that are *exchangeable* under the null (reminder: their joint distribution is unchanged by relabeling) and use the *rank* of the real statistic among simulated copies as a p-value. This section converts that idea into an e-value.

### Setup

*Intuition:* we have one real test statistic  $L_0$  and  $B$  “fake” copies  $L_1, \dots, L_B$  that look statistically identical to  $L_0$  under the null (e.g., generated by random permutations). We want to turn the comparison “is  $L_0$  unusually large?” into a valid e-value.

- $L_0$ : statistic from the real data (larger = more evidence against  $H_0$ ).
- $L_1, \dots, L_B$ :  $B$  additional statistics, exchangeable together with  $L_0$  under  $\mathcal{P}$ .
- **Example source of exchangeability:** testing independence of paired data  $(X_i, Y_i)$  via sample correlation  $L_0$ ; permute the  $X$ 's randomly  $B$  times to get  $L_1, \dots, L_B$ .

# The Soft-Rank E-value Formula

## Definition (soft-rank e-value)

First transform  $L_0, \dots, L_B$  into nonnegative scores  $R_0, \dots, R_B$  that preserve their order and exchangeability (details next slide), then define

$$E = \frac{(B+1)R_0}{\sum_{b=0}^B R_b} \quad (\text{with } E := 1 \text{ if } \sum_b R_b = 0).$$

- $R_0$ : the transformed score of the *real* data's statistic.
- $\sum_{b=0}^B R_b$ : total score across the real statistic and all  $B$  exchangeable copies.
- *Why this is an e-variable*: by exchangeability of  $R_0, \dots, R_B$  under  $\mathcal{P}$ , one can show  $\mathbb{E}^{\mathbb{P}}[R_0 \mid \sum_b R_b] = \frac{\sum_b R_b}{B+1}$ , which after rearranging gives  $\mathbb{E}^{\mathbb{P}}[E] = 1$  (exact).

## Example 1.7: A Concrete Score Transformation

### Example 1.7: one concrete transformation

For a constant  $r > 0$  and  $L_* = \min_b L_b$ :

$$R_b = \frac{\exp(rL_b) - \exp(rL_*)}{r} \geq 0.$$

- As  $r \rightarrow 0$ :  $R_b \rightarrow L_b - L_* \geq 0$  (a simple shift).
- If the  $L_b$  are already nonnegative, the simplest valid choice is  $r = 0$  with  $L_* = 0$ , i.e., **just use**  $R_b = L_b$  **directly**.

# Concrete Numbers: Soft-Rank E-value on Our Coin Data

We instantiate the abstract machinery with our own 20-flip dataset (14 heads), using the **longest run of heads** as the test statistic (a classic way to check whether a sequence looks “too streaky” to be a fair, independent coin).

## Step-by-step calculation

- **Data:** 11111101010101010111 (14 heads, 6 tails, engineered with a run of 6 heads at the start).
- **Step 1:** compute  $L_0 = \text{longest head-run in the original data} = 6$ .
- **Step 2:** draw  $B = 4$  independent random permutations of the same 20 flips; compute longest head-runs  $L_1, \dots, L_4 = 7, 5, 4, 5$ .
- **Step 3:** since all  $L_b \geq 0$ , take  $R_b = L_b$  (the  $r = 0$  choice). So  $R_0, \dots, R_4 = 6, 7, 5, 4, 5$ , summing to 27.
- **Step 4:** soft-rank e-value  $E = \frac{(4+1) \times 6}{27} = \frac{30}{27} \approx 1.111$ .
- **Step 5:** ordinary permutation p-value  $P = \frac{\#\{b : L_b \geq L_0\}}{B+1} = \frac{2}{5} = 0.4$  (namely  $b = 0, 1$  tie/exceed  $L_0 = 6$ ).
- **Check the book's inequality**  $P \leq 1/E$ : indeed  $0.4 \leq 1/1.111 = 0.9$ . ✓

# Python: Computing the Soft-Rank E-value

```
1 import numpy as np
2 def longest_run(seq, val=1):
3     best = cur = 0
4     for x in seq:
5         cur = cur + 1 if x == val else 0
6         best = max(best, cur)
7     return best
8
9 seq = [1,1,1,1,1,1,0,1,0,1,0,1,0,1,0,1,1,1] # 20-flip data, 14 heads
10 L0 = longest_run(seq)
11 rng = np.random.default_rng(11)
12 B = 4
13 Ls = [L0] + [longest_run(list(rng.permutation(seq))) for _ in range(B)]
14 R = np.array(Ls, dtype=float) # r=0 choice: R_b = L_b (nonnegative)
15 E = (B + 1) * R[0] / R.sum()
16 P = np.mean(np.array(Ls) >= L0) # ordinary permutation p-value
17 print("L_0..L_4 =", Ls, " E =", round(E,4), " P =", P, " P<=1/E:", P <= 1/E)
```

**Output:** L\_0..L\_4 = [6, 7, 5, 4, 5], E = 1.1111, P = 0.4, P<=1/E: True ( $0.4 \leq 0.9$ ) – matching the hand calculation exactly.

# Why “Soft-Rank”?

- The usual permutation p-value only looks at the *rank* of  $L_0$  among  $L_0, \dots, L_B$ :  
$$P = \sum_b \mathbf{1}\{L_b \geq L_0\} / (B + 1).$$
- After the exponential-style transform,  $1/E$  can be seen as a “smoothed” version of that rank – hence **soft-rank p-value** for  $1/E$ , and **soft-rank e-value** for  $E$  (analogous to how “soft-max” smooths the maximum via exponentiation).
- The book shows algebraically that  $P \leq 1/E$  always – the direct p-value is never larger than the soft-rank p-value, so at first glance there’s no advantage to using  $E$  for a *single* test.
- **But:** Chapters 8–11 show e-values enjoy real advantages over p-values specifically in multiple testing and building confidence intervals – advantages invisible when testing just one hypothesis.



## 1.8 Historical Context: Who Invented E-values? I

**Short answer: nobody, and everybody.** If  $\mathcal{P} = \{\mathbb{P}\}$  is simple, every optimal e-value is (or is dominated by) a likelihood ratio – and likelihood ratios are  $\sim 100$  years old, central to both frequentist and Bayesian statistics. What's new is recognizing e-values as a *general, composite-hypothesis-friendly* concept and the coordinated choice of the term “e-value” itself (agreed jointly by researchers in 2020).

The book credits five central historical figures, in chronological order:

- **Ville (1939):** brought (nonnegative) martingales to the center of probability theory; showed any measure-zero event has a nonnegative martingale exploding to  $\infty$  on it. Motivated by critiquing von Mises' theory of “collectives,” not statistical inference per se.
- **Wald (1945):** invented sequential analysis; nonnegative martingales (esp. the likelihood-ratio process) are central to his sequential probability ratio test, though framed around a single pre-specified stopping time.

## 1.8 Historical Context: Who Invented E-values? II

- **Kelly (1956):** proposed *log-optimality* – betting to maximize the expected log-growth rate of wealth in repeated favorable games, linking gambling to Shannon’s information theory. (Breiman, 1961, generalized this beyond binary outcomes.)
- **Robbins (1970, with Darling 1967):** defined confidence sequences and “power-one” tests; recognized that products of e-values yield nonnegative supermartingales, and pushed for *anytime-valid* inference rather than Wald’s fixed stopping rule.
- **Cover (1984):** developed log-optimal portfolio theory for financial trading with regret bounds against the best fixed portfolio in hindsight – a close cousin of log-optimal e-values.

**Vovk** later spearheaded the modern *game-theoretic probability* program (from 1993 onward, with Shafer and others), culminating in the term “e-value” being jointly adopted by Vovk, Shafer, Grünwald, Ramdas, and Wang around 2020.

# Recent Progress: The 2018–2020 Explosion I

A remarkable concentration of independent, aligned papers appeared on arXiv within a two-year window:

- **2018 (Howard, Ramdas, McAuliffe, Sekhon):** e-processes via families of supermartingales + Ville's inequality  $\Rightarrow$  nonparametric confidence sequences.
- **2018 (Kaufmann & Koolen):** mixture method for likelihood ratios in exponential families; sequential KL-divergence bounds.
- **2019 (Shafer):** expository case for betting-based scientific communication and log-optimality.
- **2019 (Shafer & Vovk, book):** game-theoretic foundations of probability and finance.
- **2019 (Grünwald, De Heide, Koolen):** simpler e-process definition (valid at any stopping time); reverse information projection; meta-analysis applications.

## Recent Progress: The 2018–2020 Explosion II

- **2019 (Vovk & Wang):** averaging is essentially the *only* admissible way to combine dependent e-values; e-to-p and p-to-e calibrators. This paper triggered the community-wide agreement on the term “e-value.”
- **2019 (Wasserman, Ramdas, Balakrishnan):** “universal inference” – e-values from split-likelihood-ratio statistics, valid with zero regularity conditions.
- **2020 (Vovk, Wang, Wang):** admissible p-value merging must go through e-values (via optimal transport duality).
- **2020 (Wang & Ramdas):** e-BH, an e-value analogue of Benjamini–Hochberg, controlling FDR under *arbitrary* dependence.
- **2020 (Ramdas, Ruf, Larsson, Koolen):** necessary and sufficient conditions for admissible SAVI (sequential anytime-valid inference) tools.
- **2020 (Waudby-Smith & Ramdas):** testing-by-betting for bounded-mean estimation; universal representation of nonnegative martingales.

## Unfortunate Name Clashes: Other “E-values” in the Wild

The letter “e” was popular before this book – watch out for these unrelated uses of “E-value” in other literatures:

- **BLAST E-value** (bioinformatics): expected number of chance hits in a sequence database; roughly  $E = -\log(1 - P)$  applied to a p-value – a different object entirely.
- **Sensitivity E-value** (VanderWeele & Ding, 2017, causal inference): the minimum confounding strength (on a risk-ratio scale) needed to fully explain away an observed association.
- **Bayesian E-value** (Pereira & Stern, 1999): the *epistemic* value  $ev(H \mid X)$  in the Full Bayesian Significance Test – a Bayesian evidence measure, unrelated to this book’s expectation-based definition.

Reassuringly, the terms “e-variable” and “e-process” have no known clashes – use them alongside “e-value” when precision matters.

## 1.9 Road Map of the Book I

The book is organized into three parts, building from foundations to advanced applications.

### Part I: Fundamental Concepts (Ch. 1–4)

- **Ch. 2 — Validity.** Properties of e-values under the null: Markov's inequality, calibrators, randomized tests; explicit e-variables in simple examples.
- **Ch. 3 — Efficiency.** Properties under the alternative: e-power, log-optimality, existence of powered e-values, mixture/plug-in methods.
- **Ch. 4 — Post-hoc decisions.** E-values' role when parts of the testing problem (level, decisions, or even data collection) depend on the data itself.

## 1.9 Road Map of the Book II

### Part II: Core Ideas (Ch. 5–9)

- **Ch. 5 — Universal inference:** a simple, general method for building e-variables for composite nulls/alternatives – the first known test for many “irregular” problems (though improvable).
- **Ch. 6 — Log-optimal e-variables:** a unique “numeraire” e-variable always exists for a fixed alternative, connected to the reverse information projection.
- **Ch. 7 — SAVI:** sequential anytime-valid inference – e-processes, testing by betting, optional stopping, Ville’s inequality.
- **Ch. 8 — Merging e-values:** combining via averages (arbitrary dependence) or products (independence).
- **Ch. 9 — Compound e-values:** e-Benjamini–Hochberg for false discovery rate control.

## 1.9 Road Map of the Book III

### Part III: Advanced Topics (Ch. 10–16)

- **Ch. 10:** approximate/asymptotic e- and p-values.
- **Ch. 11:** combining dependent e-values is only admissible via (weighted) averaging.
- **Ch. 12:** combining dependent p-values (via e-values and optimal transport duality).
- **Ch. 13:** e-confidence intervals; e-Benjamini–Yekutieli for false coverage rate.
- **Ch. 14:** further e-value/p-value contrasts and dualities.
- **Ch. 15:** improved (sub- $1/\alpha$ ) thresholds under shape or dependence assumptions.
- **Ch. 16:** connections to risk measures; nonparametric tests for forecasts.



## 1.9 Road Map of the Book IV

### Four cross-cutting goals

- ① **Understanding the foundations:** Chapters 2, 4, 10, 14.
- ② **Constructing good/powerful e-values:** Chapters 3, 6, 15.
- ③ **Designing inference methods:** Chapters 5, 7, 16.
- ④ **Designing multiple-testing methods:** Chapters 8, 9, 11, 12, 13.

In the first two categories, the object of interest is the e-value *itself*; in the remaining two, the e-value is a means to a specific statistical end.

# Chapter 1 — Key Takeaways

- An **e-value** is a nonnegative statistic with expectation  $\leq 1$  under the null – a valid bet against  $H_0$ ; large values are evidence against the null (opposite convention from p-values).
- E-values and p-values are **statistical cousins**: convertible via  $P = 1/E$  and  $E = \mathbf{1}\{P \leq \alpha\}/\alpha$ , but e-values uniquely support post-hoc levels, optional continuation (via simple multiplication), and unknown sampling schemes.
- The preferred optimality criterion is **e-power**  $\mathbb{E}^{\mathbb{Q}}[\log E]$ , forced by the algebra of repeatedly multiplying e-values together.
- The **betting interpretation** (Forecaster vs. Skeptic) is the intuitive engine behind the whole framework: e-variables are exactly the wealth-generating bets Forecaster would accept.
- Two worked templates recur throughout the book: the **likelihood-ratio e-value** (our coin example:  $n = 20$ , 14 heads,  $E \approx 5.18$ ) and the **soft-rank e-value** (permutation-style tests, our longest-run example:  $E \approx 1.11$ ).
- Five names anchor the history: **Ville, Wald, Kelly, Robbins, Cover** – with Vovk, Shafer, Grünwald, Ramdas, and Wang driving the modern synthesis.
- The rest of the book builds outward from these definitions: validity (Ch. 2), efficiency (Ch. 3), post-hoc decisions (Ch. 4), and onward through sequential and multiple-testing applications.